

Process modeling and parameter optimization using radial basis function neural network and genetic algorithm for laser welding of dissimilar materials

Yuewei Ai¹ · Xinyu Shao¹ · Ping Jiang¹ · Peigen Li¹ · Yang Liu¹ · Chen Yue¹

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Abstract The welded joints of dissimilar materials have been widely used in automotive, ship and space industries. The joint quality is often evaluated by weld seam geometry, microstructures and mechanical properties. To obtain the desired weld seam geometry and improve the quality of welded joints, this paper proposes a process modeling and parameter optimization method to obtain the weld seam with minimum width and desired depth of penetration for laser butt welding of dissimilar materials. During the process, Taguchi experiments are conducted on the laser welding of the low carbon steel (Q235) and stainless steel (SUS301L-HT). The experimental results are used to develop the radial basis function neural network model, and the process parameters are optimized by genetic algorithm. The proposed method is validated by a confirmation experiment. Simultaneously, the microstructures and mechanical properties of the weld seam generated from optimal process parameters are further studied by optical microscopy and tensile strength test. Compared with the unoptimized weld seam, the welding defects are eliminated in the optimized weld seam and the mechanical properties are improved. The results show that the proposed method is effective and reliable for improving the quality of welded joints in practical production.

Abbreviations

LBW	Laser butt welding
RBFNN	Radial basis function neural network
GA	Genetic algorithm
LP	Laser power
WS	Welding speed
FP	Focal position
GAP	Gap
SG	Shielding gas
DOE	Design of experiments
ANN	Artificial neural networks
PSO	Particle swarm optimization
BPNN	Back-propagation neural network
WF	Front width
HF	Front height
WB	Back width
HB	Back height
BM	Base materials
F	Focal length
BPP	The beam parameter product
WZ	Weld zone

1 Introduction

Recently, welding of dissimilar materials has been increasingly applied in automotive, ship and space industries which require certain special combination of properties [1, 2]. The welded joints of dissimilar materials can provide appropriate properties, greater flexibility and cost incurred toward costly and scarce materials. Compared with welding of similar materials, dissimilar material welding faces some new problems, such as the differences

✉ Ping Jiang
jiangping@mail.hust.edu.cn

¹ The State Key Laboratory of Digital Manufacturing Equipment and Technology, School of Mechanical Science and Engineering, Huazhong University of Science & Technology, Wuhan 430074, People's Republic of China

in thermal expansion coefficients, difficulties in heat treatment and electrochemical properties variations which can be minimized with high-density heat source. Laser welding has been widely used in industry due to its advantages of high power density, high productivity, high penetration and narrow heat affected zone, and the laser butt welding (LBW) provides a promising way for the welding of dissimilar materials. However, the hump, porosity, incomplete fusion, collapse and dripping welding defects often exist in the dissimilar materials laser welding when unsuitable process parameters are adopted. The quality of welded joints is evaluated according to the weld seam geometry, microstructures and mechanical properties [3, 4]. It is observed that the welding process parameters, such as the laser power (LP), welding speed (WS), focal position (FP), gap (GAP) and shielding gas (GS), have great influence on the weld seam geometry and hence the quality of welded joints. How to select proper process parameters for the LBW of dissimilar materials to improve the quality of weld seam has become an important challenge.

The relationship between the welding process parameters and weld seam geometry is non-linear and complicated, especially the extent of contribution of actual process parameters towards the desired result is difficult to be identified. Choosing the proper process parameters for the desired seam geometry and high-quality welded joints is always a puzzling problem in practical production. Traditionally, the welding process condition is usually determined based on the welders' experiences, charts and handbooks [5, 6]. Selecting a set of parameters by the trial-and-error method may yield fairly good results, but it is hard to ensure an optimal weld seam, and particularly the method is not applicable to the new welding process. To overcome these existing problems and obtain the satisfactory welding parameters without consuming time, materials and labor effort, the correlation between input variables with output results is required to be quantified by mathematical models. The design of experiments (DOE) and statistical techniques are the frequently used methods in determining the welding process parameters. Tarng et al. [7] adopted the Taguchi method to obtain the set of optimal welding process parameters. The effect of welding process parameter on weld seam geometry is analyzed and the optimal parameter settings are determined. Dongxia et al. [8] used Taguchi approach as a statistical design of experimental technique for optimizing the selected welding parameters. The established model showed the effect of welding process parameters on the welding quality decreased in the order of WS, wire feed rate and LP. The optimal combination of welding parameters was also determined. Due to the inadequacy and inefficiency of the proposed mathematical models above, their applications

are limited. Artificial neural networks (ANN) as the models with good predicting performance have been introduced into the welding process. Dutta et al. [9] compared regression analysis and a neural network algorithm in modeling the weld seam width and penetration depth of tungsten inert gas welding and proved that the predicted performance of neural network is better than the regression model. Katherasan et al. [10] researched simulation and parameter optimization of flux-cored arc welding using ANN and particle swarm optimization (PSO) algorithm. The maximization of depth of penetration, minimization of seam width and minimization of reinforcement were obtained. The neural network approach was found to be good in predicting outputs from a given set of input parameters. Intelligence algorithms have been used for the nonlinear process optimization such as the welding process [9–11]. Park et al. [11] studied the process modeling and parameter optimization using back-propagation neural network (BPNN) and genetic algorithm (GA) for aluminum laser welding. Based on the established ANN model, GA was taken to optimize the LP, WS and wire feed rate. The optimal results demonstrated in good agreement with the actual values from the experiments. Sathiya et al. [12] reported the weld seam geometry and mechanical property of the laser welding affected by input parameters (LP, WS, FP and SG) using factorial design, ANN and GA. From the above studies, it is showed that the artificial intelligence method was capable of locating optimum conditions for the welding of similar materials.

On the other hand, the welding of dissimilar materials is much more challenging than that of similar materials due to metallurgical incompatibility, thermal mismatch, etc. The welding process accompanies with many complex natural phenomena and the affecting factors may not be understood fully. There are a few preliminary studies focused on the weld seam characteristic of dissimilar materials at the certain welding process parameters. Meshram et al. [13] reported the microstructures and tensile properties of the weld region in friction welding of dissimilar pure metals. Paventhan et al. [14] studied fatigue behavior of friction-welded joints of the medium carbon steel and austenitic stainless steel. Satyanarayana et al. [1] obtained sound welds at certain combinations of weld process parameters only. To further study the influence of the process parameters, DOE and statistical methods are gradually introduced into the welding of dissimilar materials by some researchers. Anawa et al. [2] used the Taguchi method to optimize welding pool of dissimilar laser-welded components. Ruggiero et al. [15] reported weld seam profile and costs optimization of the CO₂ dissimilar laser welding process of the low carbon steel and austenitic steel AISI316 using response surface methodology. Koilraj et al. [16] optimized the process parameters in

the friction stir welding of dissimilar aluminum alloys AA2219–AA5083 using Taguchi technique. However, the problems in dissimilar materials welding have not been solved well by the conventional methods. The artificial intelligence methods have not been used in the laser welding of dissimilar materials. Moreover, most of the previous studies get the optimal seam geometry only, but the corresponding microstructures and mechanical property of the weld seam obtained from the optimal process parameters were not further analyzed. There may be some invisible defects existing in the optimal weld seam, which reduce the quality of welded joints directly.

Therefore, this paper presents an artificial intelligence method to optimize the weld seam geometry for the low carbon steel (Q235) and stainless steel (SUS301L-HT) laser butt welding (LBW) by using radial basis function neural network (RBFNN) and GA. Based on the established RBFNN model, the process parameters are optimized by GA obtain the weld seam with minimum seam width and desired depth of penetration. The input process parameters include the LP, WS, FP and GAP, whereas the front width (WF), front height (HF), back width (WB) and back height (HB) of the weld seam are considered as the output responses. The four-factor and five-level experiment is designed by Taguchi. The optimal results are verified experimentally. Simultaneously, to further confirm the quality of weld seam produced under the optimal process condition, the microstructures and mechanical properties are analyzed. The comparison between the weld seams generated by optimized parameters and random parameters demonstrate that the geometric optimization by GA with respect to the geometry is also beneficial regarding the defect elimination and mechanical properties improvement.

The structure of the paper is as follows. Section 2 introduces the process parameters for LBW of dissimilar materials, welding system, experimental details on the materials and DOE. Section 3 develops the RBFNN to establish the relationship between the input parameters and output results and illustrates GA methodology for process parameters optimization. Section 4 describes the confirmation of the optimal results, and quality analysis and comparison of the optimal weld seam on the defects, microstructures and mechanical properties. The conclusions of the present study are outlined in Sect. 5.

2 Experiment procedure

2.1 Materials

The low carbon steel (Q235) is cheap and has excellent welding performance. The stainless steel provides good corrosion resistance. The welded joints of the low carbon

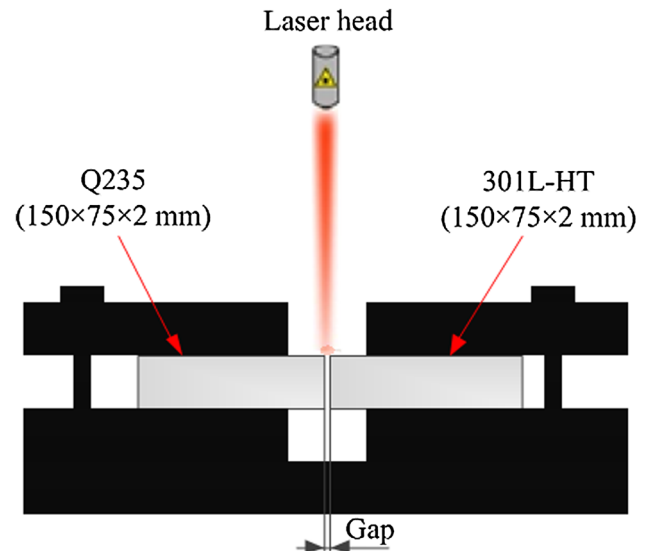


Fig. 1 Schematic diagram of the welding process

Table 1 Chemical composition of low carbon steel (Q235) (wt%)

Material	C	P	S	Si	Mn
Q235	0.014	0.017	0.038	0.10	0.038

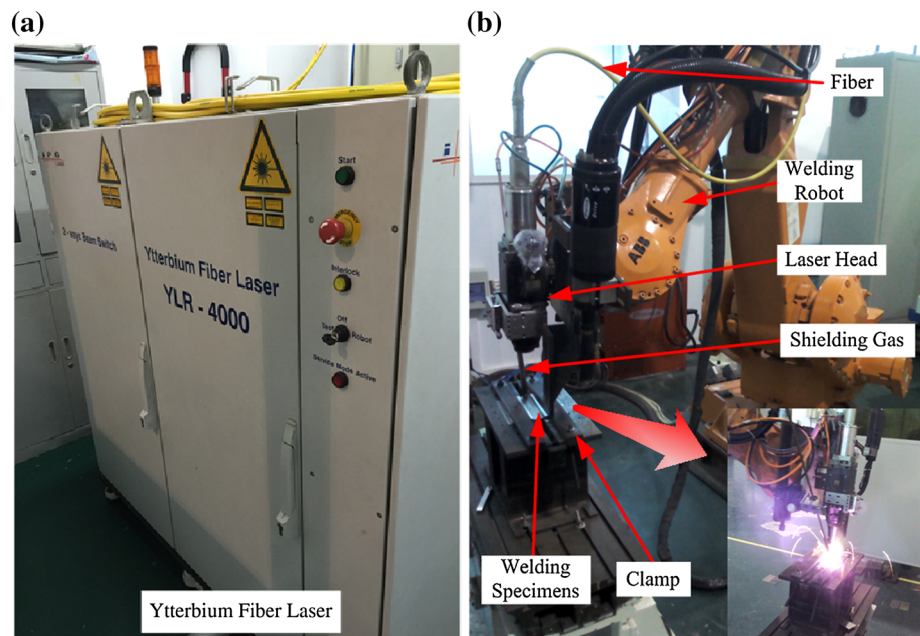
steel (Q235) and stainless steel (SUS301L-HT) can preserve the corrosion resistance at a low cost. So, the low carbon steel (Q235) and stainless steel (SUS301L-HT) were selected in the experiments as the base materials (BM) for LBW, as shown in Fig. 1. The compositions of the low carbon steel and stainless steel are presented in Tables 1 and 2. They were cut into small plates as the welding specimens, and their dimensions were $150 \times 75 \times 2 \text{ mm}^3$. To control GAP of the butt welding strictly, the butt surface of two types of specimens were milled in the milling machine. All surfaces of the specimens were cleaned by acetone to protect the weld seam from disturbing with oil pollution and oxide film.

2.2 Laser butt welding system

The LBW system in this study is shown in Fig. 2. The laser is ytterbium fiber laser (IPG YLR-4000) with a wavelength of $1.07 \mu\text{m}$. The spot diameter is focused into 0.3 mm with lens of 250 mm focal length (F). The beam parameter product (BPP) is 6.3 mm-mrad. In the whole welding experiments, continuous wave fiber laser was used. The welding head was equipped with coaxial air blowing protection device and grasped on a welding robot, ABB IRC5 M2004. The controlling system was used to set the welding process parameters, LP, WS and FP included. The last parameter, GAP, was controlled by the feeler gauge.

Table 2 Chemical composition of stainless steel (SUS301L-HT) (wt%)

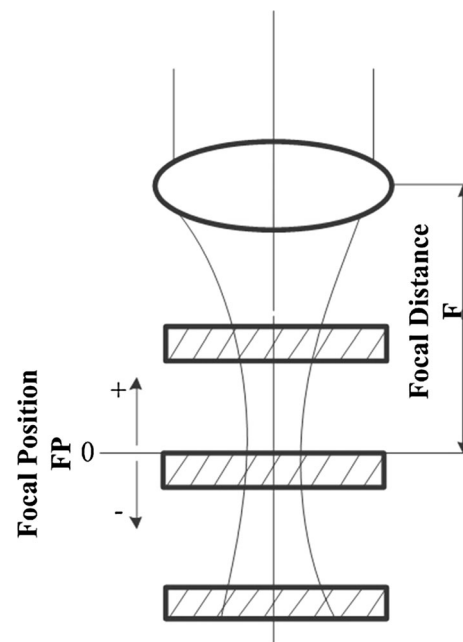
Material	C	P	S	Si	Mn	Cr	Ni	N
SUS301L-HT	0.018	0.027	0.002	0.47	1.24	17.75	7.50	0.10

Fig. 2 Laser butt welding system. **a** Fiber laser; **b** welding system

During the welding process, the side-blown Argon SG was supplied laterally at a flow rate of 30 L/min. Since the LBW considered here was autogenous welding, the weld wire equipment was not used.

2.3 Input process parameters and output responses

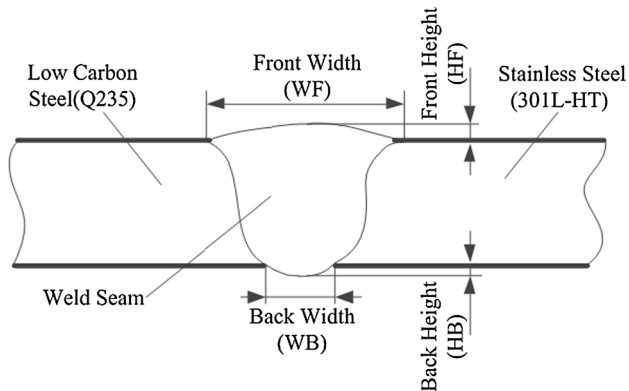
According to the practical experience and previous research [2, 15], it is observed that the process parameters play an important role in the weld seam geometry and the quality of welded joints. The process parameters include LP, WS, FP, GAP and SG in the LBW. In the current research, the SG and gas flow rate were kept constant. The main process parameters, LP, WS, FP and GAP, were studied. Low LP and high WS are not good for flowing of the molten metal, which will lead to the defects formation of incomplete fusion, hump, porosity, while wide GAP and big FP will reduce the laser absorptivity and cause the dripping of the molten metal. The FP is shown in Fig. 3. The FP is distance between the focal point and the top surface of the welding specimen. When the focal point is at the top surface of the welding specimen, the FP is defined as 0, and below the top surface, the value of FP is positive, otherwise, is negative. The range of each process parameter is required to be limited to obtain the weld seam with desired geometry and without defects. In this study, the

**Fig. 3** Illustration of the focal position

upper and lower bound and levels of each process parameter were presented in Table 3. The LP was adjusted from 1500 to 3500 W, the WS was set up between 2.5 and

Table 3 Welding process parameters and their levels

Parameter	Unit	Notation	Factor levels				
			1	2	3	4	5
Laser power	W	LP	1500	2000	2500	3000	3500
Welding speed	m/min	WS	2.50	2.75	3.00	3.25	3.50
Focal position	mm	FP	−3.0	−2.0	−1.0	0.0	1.0
Gap	mm	GAP	0.00	0.02	0.04	0.06	0.08

**Fig. 4** Butt welding seam geometry

3.5 m/min, the FP was selected from −3.0 to 1.0 mm and GAP was ranged from 0 to 0.08 mm.

The weld seam is generally characterized by geometrical features, seam width, height and penetration. In the LBW, the evaluation parameters of the weld seam geometry, WF, HF, WB and HB, were selected as the output responses which were illustrated in Fig. 4.

2.4 Design of experiments

According to the previous discussion, there are four main process parameters considered in this study and the limited range of each parameter was divided into five levels with equal interval, as shown in Table 3. So, the present problem is converted into a four-factor five-level problem. The DOE was made by Taguchi in a statistical software, MINITAB 16. Table 4 shows the generated four-factor and five-level matrix as Taguchi L_{25} . Thus, 25 groups of experiments were conducted by the designed matrix to avoid the results to be disturbed by systematic errors. LP was set in the fiber laser IPG YLR-4000, WS and FP were adjusted in the ABB IRC5 M2004 welding robot, and the GAP of the butt welding was measured by a feeler gauge.

2.5 The results of experiment

To obtain the clear metallographic figure under the optical microscope, the small specimens ($15 \times 15 \times 2 \text{ mm}^3$) were cut from the weld seam. Then, the specimens were polished

Table 4 Taguchi L_{25} design matrix

No.	LP (W)	WS (m/min)	FP (mm)	GAP (mm)
1	1500	2.50	−3	0.00
2	1500	2.75	−2	0.02
3	1500	3.00	−1	0.04
4	1500	3.25	0	0.06
5	1500	3.50	1	0.08
6	2000	2.50	−2	0.04
7	2000	2.75	−1	0.06
8	2000	3.00	0	0.08
9	2000	3.25	1	0.00
10	2000	3.50	−3	0.02
11	2500	2.50	−1	0.08
12	2500	2.75	0	0.00
13	2500	3.00	1	0.02
14	2500	3.25	−3	0.04
15	2500	3.50	−2	0.06
16	3000	2.50	0	0.02
17	3000	2.75	1	0.04
18	3000	3.00	−3	0.06
19	3000	3.25	−2	0.08
20	3000	3.50	−1	0.00
21	3500	2.50	1	0.06
22	3500	2.75	−3	0.08
23	3500	3.00	−2	0.00
24	3500	3.25	−1	0.02
25	3500	3.50	0	0.04

by 400, 800, 1000, 1200, 1500, 2000 and 2500 grits SiC abrasive papers, respectively. After that, they were further polished with lapping paste. Finally, the polished specimens were corroded using 4 % nitric acid alcohol to display the weld seam geometry clearly, and the metallographic specimens were obtained. The metallographic figure for each specimen was taken by microscope. After getting the metallographic figure of weld seam, the seam geometry of the LBW was measured by the measurement software, CSM1. The observed results of the seam geometry were presented in Table 5. For the HF, the negative values mean that the upper bound of the weld seam is lower than the upper edge of the BM, while the negative values for the HB denote

Table 5 Experimental results

No.	LP (W)	WS (m/min)	FP (mm)	GAP (mm)	WF (μm)	HF (μm)	WB (μm)	HB (μm)
1	1500	2.5	-3	0	1305	31	0	-355
2	1500	2.75	-2	0.02	1135	-40	20	-275
3	1500	3	-1	0.04	1023	-100	40	-215
4	1500	3.25	0	0.06	1009	-193	60	-143
5	1500	3.5	1	0.08	847	-183	80	-175
6	2000	2.5	-2	0.04	1488	-10	717	52
7	2000	2.75	-1	0.06	1239	-84	585	234
8	2000	3	0	0.08	1161	-106	423	82
9	2000	3.25	1	0	1005	-42	0	-383
10	2000	3.5	-3	0.02	1368	138	456	-381
11	2500	2.5	-1	0.08	1122	-19	920	-175
12	2500	2.75	0	0	1400	27	758	110
13	2500	3	1	0.02	1572	101	513	216
14	2500	3.25	-3	0.04	1174	-83	863	-20
15	2500	3.5	-2	0.06	1079	-83	762	88
16	3000	2.5	0	0.02	1255	-65	1133	80
17	3000	2.75	1	0.04	1452	-2	832	76
18	3000	3	-3	0.06	1523	-134	873	-6
19	3000	3.25	-2	0.08	1105	-148	983	-109
20	3000	3.5	-1	0	990	-134	845	-236
21	3500	2.5	1	0.06	1192	-119	1059	-151
22	3500	2.75	-3	0.08	1291	44	1169	-92
23	3500	3	-2	0	1368	-31	1289	221
24	3500	3.25	-1	0.02	1312	-108	1394	199
25	3500	3.5	0	0.04	1545	-103	1379	66

incomplete penetration or the bottom of the weld seam is higher than the bottom of the BM.

3 Optimization methodology and implementation

RBFNN is suitable for predicting nonlinear problems, and GA has the capacity to optimize parameters. In this study, the integrated method (RBFNN-GA) is presented to optimize process parameters of the LBW of dissimilar materials. The experimental results are fed into the RBFNN to establish the relationship between the input process parameters and output responses. The predicted results from the RBFNN are optimized by GA with the objective function. The flowchart of the optimization methodology is shown in Fig. 5. The procedure in detail is described as follows:

1. Experimental results. The experiments were carried out according to the Taguchi L_{25} matrix. The experimental results were recorded in Table 5.
2. Simulation using RBFNN. The RBFNN is constructed and initialized based on the input process parameters and the identified output responses. Then, the

experimental results are used to train and test the RBFNN to improve the predicting accuracy. The output responses predicted from the trained network are taken for the process parameters optimizing.

3. Optimization of the process parameters using GA. In this part, the predicted outputs are optimized by GA with the minimum weld area and desired depth of penetration, and the optimal process parameters are obtained in the optimal solution.

3.1 Simulation using RBFNN

3.1.1 RBFNN

RBFNN is proposed by Moody and Darken [17]. RBFNN has been used for a wide range of application, such as function approximation and pattern recognition [18–20], due to the advantages that it can approximate any regular function [18], its training is faster than that of a multilayer perceptron [19], and it is best suited for learning to approximate continuous or piecewise continuous real-valued mappings [21, 22]. An RBFNN is a three-layer feed-forward network that consists of one input layer, one

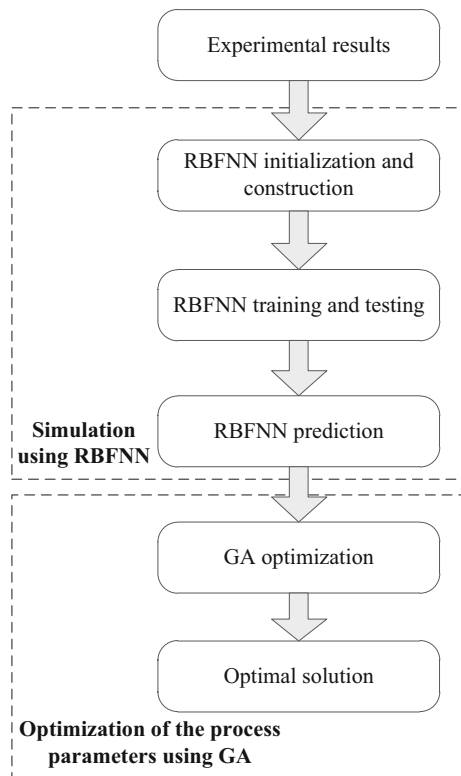


Fig. 5 Flowchart of the optimization methodology (RBFNN-GA)

middle layer and one output layer. Every layer is connected to each other by the basic unit of RBFNN, neuron, associated with weight factors. The construct of RBFNN can be summarized as follows [17–22].

1. **RBFNN initialization:** Before RBFNN constructing, the initializing operation is necessary. Firstly, the number of neuron of every layer should be determined. According to the number of input parameters and output responses in the experiments, the neuron number of input layer and output layer are set up, respectively. The middle layer will be decided by selecting n samples randomly from the training patterns or using clustering algorithms such as K-means or Self-Organizing Feature Map. Then, radial basis function and the diffusion rate of the radial basis function are selected and the RBFNN initialization is accomplished.
2. **RBFNN constructing:** Each input neuron corresponds to a component of an input vector $\mathbf{X}$. The middle layer includes m neurons and one bias neuron. The input neurons are fully connected to the middle layer neurons except the bias neuron. The middle layer neurons y_j are computed by the radial basis function which is preferred to select the Gaussian function due to the fact that it is factorable. Hence

$$\mathbf{X} = [x_1, x_2, x_3, \dots, x_n]^T \quad (1)$$

$$y_j = \begin{cases} \exp \left[-\frac{\|\mathbf{X} - \mathbf{C}_j\|^2}{2b_j^2} \right] & j = 1, 2, \dots, m \\ 1 & j = 0 \text{ (bias neuron)} \end{cases} \quad (2)$$

In Eq. (2), $\|\cdot\|$ indicates the Euclidean distance. The j th center vector of the network is:

$$\mathbf{C}_j = [c_{j1}, c_{j2}, c_{j3}, \dots, c_{jn}]^T, \quad j = 1, 2, \dots, n \quad (3)$$

The $\mathbf{C}_j$ denotes the weight vector between the input layer and the j th middle layer neuron. The radial width vector of the middle layer is:

$$\mathbf{B} = [b_1, b_2, b_3, \dots, b_n]^T \quad (4)$$

b_j is the width of the j th neuron in the middle layer and all of them are positive. The weight vector between the output layer and the middle layer is:

$$\mathbf{W}_k = [w_{k1}, w_{k2}, w_{k3}, \dots, w_{km}]^T, \quad k = 1, 2, \dots, N \quad (5)$$

The output layer consists of k neurons which correspond to the identified objectives and it is fully connected to the middle layer.

Each neuron in output layer is obtained from computing a linear weighted sum of the outputs of the middle layer as follows:

$$z_j = \sum_{i=0}^m y_i w_{ij}, \quad j = 1, 2, \dots, N \quad (6)$$

After the RBFNN constructed, the collected experiment data can be fed into the RBFNN for training and testing.

3.1.2 RBFNN parameter settings

Before the LBW process parameter prediction, the parameters of RBFNN should be set. The RBFNN parameter settings consist of the neuron number of input layer and output layer, and the diffusion rate of the radial basis function, spread. In this study, there are four input parameters, including LP, WS, FP and GAP, and four output parameters, WF, HF, WB and HB. So, the neuron number of both the input layer and output layer is four. The spread is set as 2.5 using the trial-and-error method.

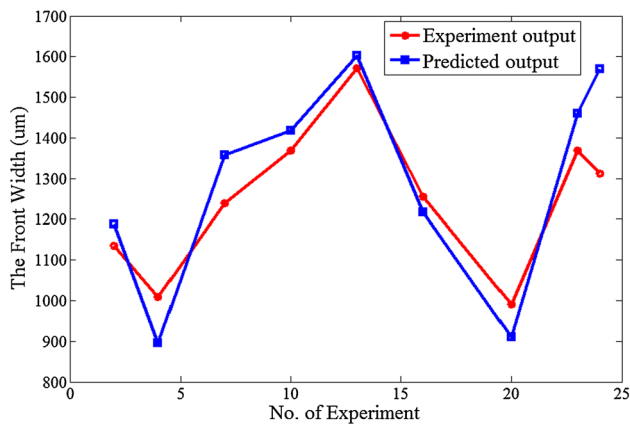
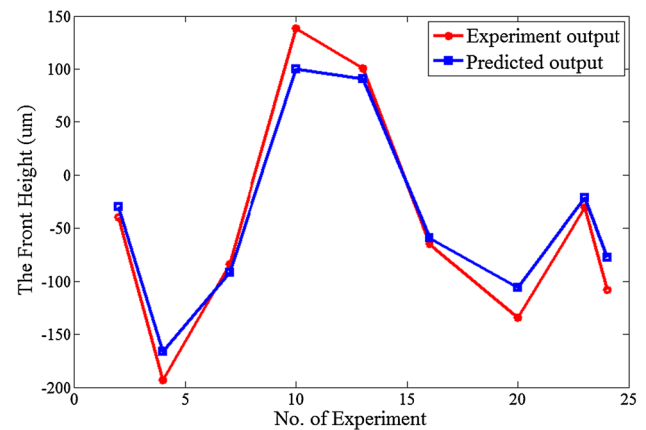
3.1.3 The training and testing of the RBFNN

The program was written and run in MATLAB R2013a. About two-thirds of the experimental data were fed into the RBFNN for training. In this part, the experiments, 1, 3, 5, 6, 8, 9, 11, 12, 14, 15, 17, 18, 19, 21, 22, 25, are randomly

Table 6 Comparison of experimental and RBFNN-simulated results of the front width, front height, back width and back height

No.	WF (μm)			HF (μm)			WB (μm)			HB (μm)		
	EV	RBFV	Err%	EV	RBFV	Err%	EV	RBFV	Err%	EV	RBFV	Err%
2	1135	1178.5456	-3.84	-40	-32.0865	19.78	20	26.3355	-31.68	-275	-266.4171	3.12
4	1009	900.8221	10.72	-193	-161.8525	16.14	60	74.8461	-24.74	-143	-157.2681	-9.98
7	1239	1347.5204	-8.76	-84	-92.9781	-10.69	585	698.1732	-19.35	234	247.0834	-5.59
10	1368	1468.9210	-7.38	138	112.0393	18.81	456	371.1571	18.61	-381	-503.4504	-32.14
13	1572	1663.7527	-5.84	101	96.3179	4.64	513	479.1135	6.61	216	178.9920	17.13
16	1255	1223.6640	2.50	-65	-70.5408	-8.52	1133	986.4601	12.93	80	68.5489	14.31
20	990	903.6687	8.72	-134	-111.2840	16.95	845	958.7602	-13.46	-236	-207.2128	12.20
23	1368	1456.7271	-6.49	-31	-35.0852	-13.18	1289	1474.7343	-14.41	221	177.6467	19.62
24	1312	1559.8789	-18.89	-108	-89.4630	17.16	1394	1551.3987	-11.29	199	232.4407	-16.80
STD DEV			8.70			13.14			16.35			16.62

EV experimental values, RBFV output values of RBFNN, Err error = $(EV - RBFV)/EV \times 100 \%$, STD DEV standard deviation

**Fig. 6** Comparison of experimental and RBFNN-simulated results of the front width**Fig. 7** Comparison of experimental and RBFNN-simulated results of the front height

selected and taken to train the RBFNN. The remaining experiments, 2, 4, 7, 10, 13, 16, 20, 23, 24, are used for testing the trained RBFNN. After the RBFNN is trained, the simulated output responses are generated by inserting the input variables to the network. To validate the accuracy of the trained RBFNN, the results obtained from simulation and experiment are compared and presented in Table 6 and plotted in Figs. 6, 7, 8 and 9. As shown in this table and figures as well, the average errors of the four output responses are relatively small and every measured parameter is accurate in microns. The standard deviation of error for each response is $<18 \%$, which demonstrates that the performance of RBFNN is evenly distributed. So, these errors are acceptable in the welding process.

These errors in the test experiments may be due to the laser intensity fluctuation, clamping error, distribution

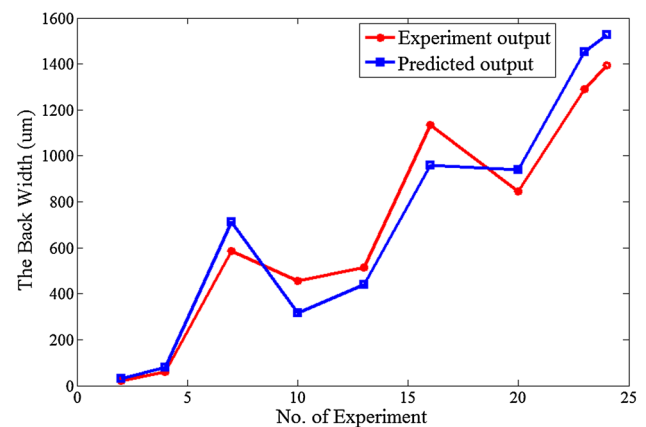
**Fig. 8** Comparison of experimental and RBFNN-simulated results of the back width

Fig. 9 Comparison of experimental and RBFNN-simulated results of the back height

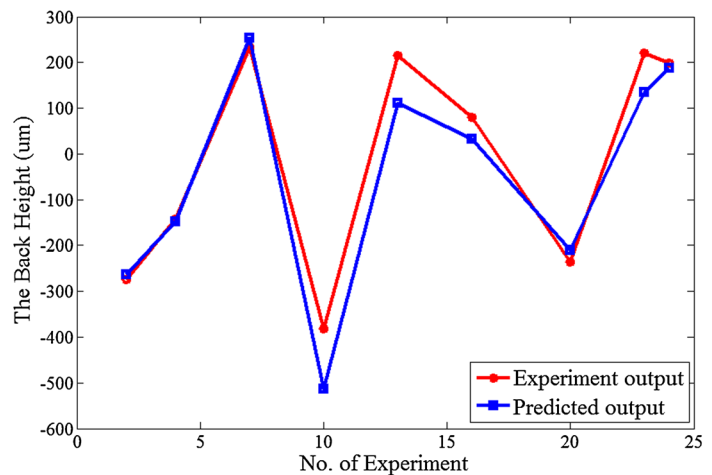


Table 7 Comparison of the prediction results with other published papers using neural network models

Paper	Welding method	Joint type	Prediction model	Maximum error (%)	Number of exceeded 20 %		Standard deviation of error (%)
					Number	Percentage (%)	
Nagesh [23]	TIG	Butt joint	RBFNN	247.76	7/60	11.7	40.99
Chokkalingham [24]	TIG	Butt joint	ANN	-72.45	3/20	15.0	22.51
Manikya [25]	GMA	Butt joint	BPNN	29.13	3/12	25.0	13.46
This paper	LBW	Butt joint	RBFNN	-32.14	5/36	8.3	15.13

differences of material structure, environmental influences and other random factors which are difficult to be controlled in the experiments. Compared with the maximum error of 247.76, 72.45 and 29.13 % in other published works [23–25], the predicted error of the RBFNN model in this paper is also relatively small, as shown in Table 7. From the table, the error exceeding 20 % of the LBW seam shape is 8.3 % and the precision is better than other references (25, 15 and 11.7 %). The standard deviation of errors is 15.13 % which is smaller than 40.99 and 22.51 %. Therefore, the predicted result demonstrates good conformability to the experimental result and can be used for optimization of welding process parameters.

3.2 Optimization of the process parameters using GA

3.2.1 Genetic algorithm

GA is a type of random search and optimization algorithm evolved from the mechanics of natural selection and genetics, and conceived by Holland [26]. It has been successfully used in a wide variety of fields such as mixed model assembly, design optimization, process parameters optimization, and so on [27–32]. The GA operates over a set of individuals which is generated in the search space randomly. Each individual represents a

possible solution of the problem. Among the search space, the GA considers a set of possible solutions simultaneously according to the fitness value of each individual obtained from the fitness function. Genetic algorithms (Fig. 10) aim at finding the most fitness objective over generation evolution through the three genetic operators: selection, crossover and mutation which are summarized as follows [11, 26].

1. **Selection** This operator is to select more fit members in the individuals based on their fitness values to produce the next generation. The individual will be selected as the parent of the next generation if it has a high fitness value. In this study, the roulette wheel selection strategy is used to imitate Darwin's survival of the fittest theory. Based on the concept of selection probability for each individual proportional to the fitness value, the probability P_i of i th individual to be selected is formulated:

$$P_i = \frac{f_i}{\sum_{j=1}^n f_j} \quad (7)$$

Where n is the size of population. f_i is the fitness value of i th individual. Each generation individuals' parents are selected as the above principle and stored in a mating pool.

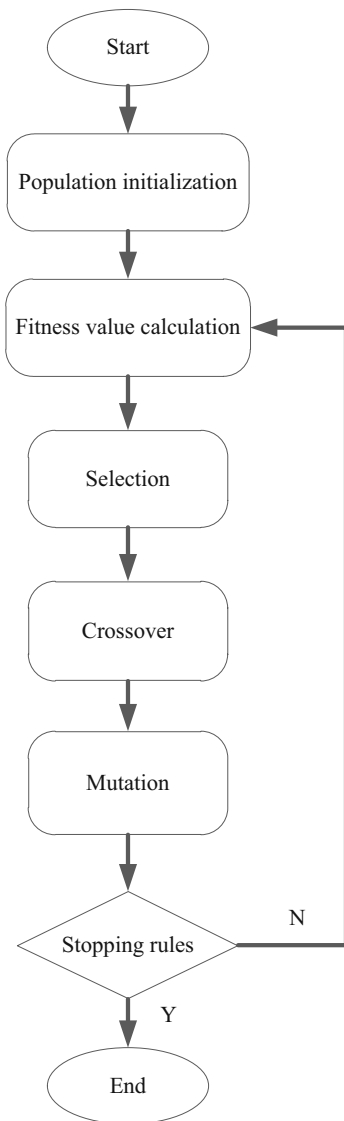


Fig. 10 Flowchart of GA

2. **Crossover** In crossover, two parents are randomly selected from the mating pool and the crossover position of the chromosomes in both parents is determined. Then, the portions of chromosomes in the crossover position are exchanged and two offspring individuals are created. Nonetheless, the crossover operation will not occur with the whole chromosomes, and it is limited by a low value, the crossover rate which is often set in the range from 0.4 to 0.99.
3. **Mutation** The role of mutation is to keep the diversity of population and generate more excellent offspring so that it can avoid producing a local optimum instead of the global optimal solution. In the operator, random change of the chromosomes happens in an individual

by selecting some symbol in a chromosome. The mutation rate is set very low as that of the biological universe to avoid losing good individuals. Simultaneously, the characteristics not existing in individual will be created. The mutation rate is often selected from 0.0001 to 0.1.

3.2.2 GA parameter settings

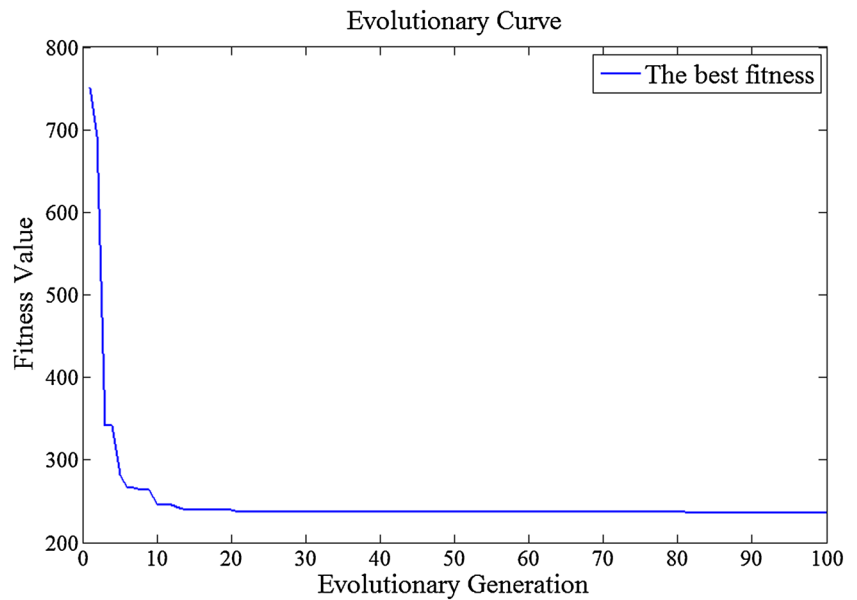
The parameters of GA are set to generate an initial population randomly and find the most fitness objective by selecting fitness function. The parameters consist of population size, the evolutionary generations, crossover rate, mutation rate and the range of the process parameters of the LBW. To obtain a better optimization result, the population size, evolutionary generations, crossover rate and mutation rate were set as 60, 100, 0.7 and 0.1, respectively, and the range of LP, WS, FP and GAP were set in the same as in the Taguchi DOE (shown in Table 5). The LP was adjusted from 1500 to 3500 W, the WS was set up between 2.5 and 3.5 m/min, the FP was selected from −3.0 to 1.0 mm and GAP ranged from 0 to 0.08 mm.

3.2.3 The optimization results

The optimization of the parameters was done in MATLAB R2013a software by using GA. The quality of the weld seam is greatly determined by the weld seam geometry which includes WF, HF, WB and HB. The desired weld seam is with minimum seam width and the depth of penetration is equal to the BM. Therefore, an attempt is made to minimize the four output responses, the values of WF, HF, WB and HB. Since there are some negative values existing in the HF and HB, all of the values of the four output responses are normalized to their absolute values in the optimization process. The objective function is defined as a sum of WF, HF, WB and HB with different weight factors, as formulated in Eq. (8).

$$F = \omega_{WF}|WF| + \omega_{HF}|HF| + \omega_{WB}|WB| + \omega_{HB}|HB| \quad (8)$$

Where ω_{WF} , ω_{HF} , ω_{WB} and ω_{HB} are the weight factors for WF, HF, WB and HB. In this study, their values are set as 0.15, 0.35, 0.10 and 0.40, respectively [10]. Through GA, the process parameters are optimized. The evolutionary curve of the best fitness value of F is shown in Fig. 11. The optimal values of the process parameters, LP, WS, FP and GAP, are 2500.974 W, 3.491 m/min, −0.856 and 0.021 mm, respectively, as presented in Table 8. The obtained optimum F is 232.041 μm and the optimized values are in line with the existing literatures [10, 32].

Fig. 11 Evolutionary curve of parameter using RBFNN-GA in LBW**Table 8** Optimal results by RBFNN-GA

F (μm)	WF (μm)	HF (μm)	WB (μm)	HB (μm)	LP (W)	WS (m/min)	FP (mm)	GAP (mm)
232.041	1178.378	-1.851	530.741	-3.897	2500.974	3.491	-0.856	0.021

Table 9 Confirmation test results

Algorithm	F (μm)	WF (μm)	HF (μm)	WB (μm)	HB (μm)	LP (W)	WS (m/min)	FP (mm)	GAP (mm)
Optimal solution (RBFNN-GA)	232.041	1178.378	-1.851	530.741	3.897	2500.974	3.491	-0.856	0.021
Feasible solution	235.35	1175	-2	568	4	2500	3.5	-1	0.02
Err (%)	-1.43	0.29	-8.05	-7.02	-2.64	-	-	-	-

$$\text{Err error} = [(\text{Optimal solution}) - (\text{Feasible solution})]/(\text{Optimal solution}) \times 100 \%$$

4 Results and discussion

4.1 Confirmation tests

To verify the optimal results, the confirmation test was carried out. Taking the actual feasibility of LBW process into consideration, the feasible solution which is the near optimum values of RBFNN-GA (shown in Table 8) was selected and presented in Table 9. The dimensions of the weld seam are shown in Fig. 12 which is obtained from the feasible solution and measured by the same measuring software as the one used in the experimental procedure, the CSM1 software. From Table 9, the optimal F is 232.041 and the corresponding feasible solution value is 235.35, and the error between them is -1.43 %. The values of the dimensions of the weld seam of the feasible solution are also very close to the optimal solution values. It is clearly

seen that the weld seam geometry from the feasible solution is in good agreement with the optimal results of weld seam geometry, and the error is negligibly small. Therefore, the weld seam geometry of the final optimization result is found to be reliable and acceptable.

The weld seams generated by some random parameters is shown in Fig. 13. From the Fig. 13, there is the porosity, incomplete fusion, collapse and dripping welding defects existing in the weld seams, which affect the quality of welded joints greatly. Compared with the unoptimized weld seam, the optimal weld seam as shown in Fig. 12 is achieved with minimum width and desired depth of penetration and the defects are eliminated well without consuming time, materials and labor effort. Therefore, the geometric optimization of the weld seam is beneficial for the improvement of welding quality and essential for the LBW of dissimilar materials process.

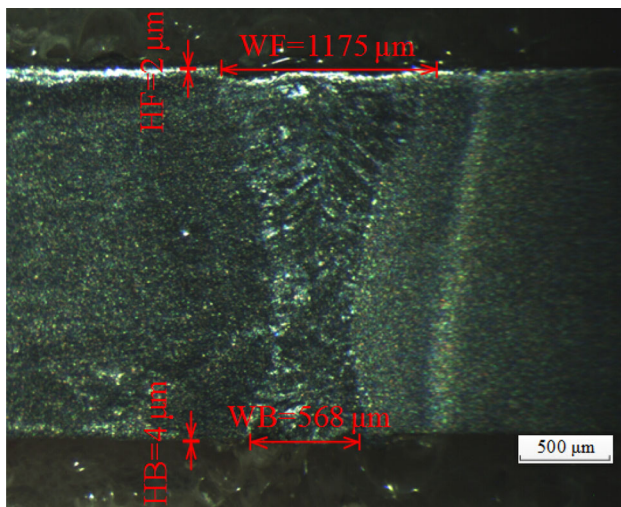


Fig. 12 The cross section of the weld seam generated by optimized parameters

4.2 Microstructures

The macro appearance can be obtained by observing the weld seam shape, and the optimal weld seam geometry is confirmed in Sect. 4.1. However, the quality of the weld seam is also determined by the microstructures and mechanical properties. Some invisible defects may exist in the optimal weld seam which also leads to reduce the quality of welded joints directly. To further confirm the quality of the weld seam produced under the optimal process condition, the microstructures and tensile strength in detail are discussed in Sects. 4.2 and 4.3.

In this study, the BM microstructures, the low carbon steel (Q235) and stainless steel (SUS301L-HT) are presented in Fig. 14. The grain boundaries are exhibited clearly. Figure 15a–d shows the microstructures of the weld zone (WZ) and interface between WZ and BM. From the figures, there are columnar structure and equiaxed grains included in the WZ microstructure. The distribution

of the columnar structure and equiaxed grains and the direction of the columnar grain are varied. The WZ, BM and fusion line can be distinguished easily in Fig. 15c, d. This type of microstructure is beneficial for the improvement of the quality of butt welded joints. Consequently, the obtained microstructure of the optimal weld seam is found to meet the requirement.

4.3 Tensile strength

Finally, the mechanical property of the optimal weld seam is detected in the tensile test. The whole tensile process was conducted on a tensile tester (TRAPEZIUM2, SHIMADZU, Japan). The crosshead speed was 1 mm/min. To avoid the system error, three tensile test specimens cut from the same butt welding plate were prepared. The tensile strength of the low carbon steel (Q235) is 381 Mpa, and the tensile strength of the stainless steel (SUS301L-HT) is 1120 Mpa. When the tensile strength of the weld seam is higher than that of the BM, the fracture will occur at the BM. Otherwise, The fractured position will be in the weld seam. During the tensile test, the prepared test specimens were installed on the tensile machine. With the increase in the loading force, the test specimens are elongated gradually and fractured ultimately. The testing results of the three test specimens are 380.572, 379.380 and 379.105 Mpa, respectively. The standard deviation for tensile strength is as small as 0.6 Mpa which denotes high consistency. The corresponding average tensile strength is 379.685 Mpa. The tensile-tested specimens are shown in Fig. 16. It is seen that the specimens from optimal process parameters are all fractured at the BM, the low carbon steel (Q235), a distance away from the weld seam. Therefore, the tensile strength of the weld seam is higher than that of the BM of the low carbon steel (Q235). The mechanical property of the obtained weld seam from the optimal process parameters meets the required performance.

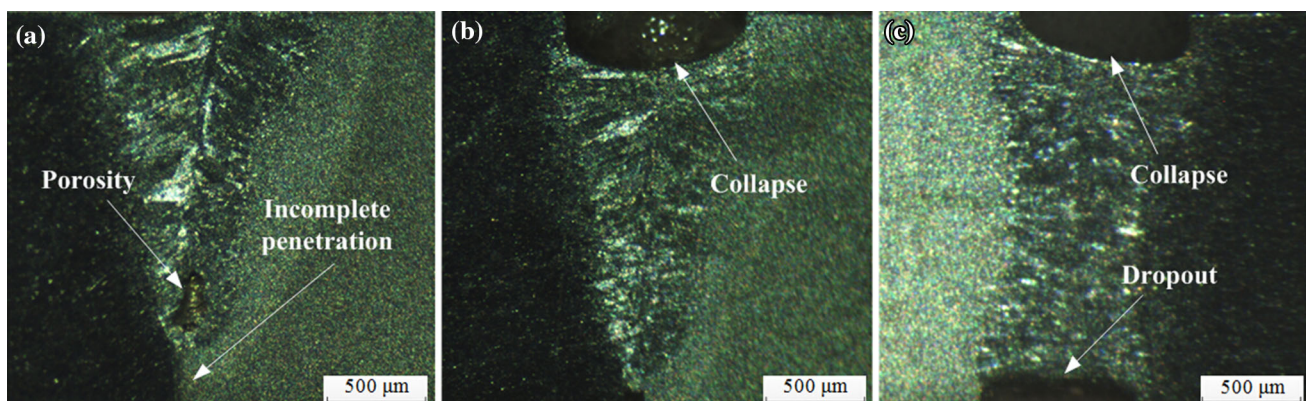


Fig. 13 Cross section of the weld seam generated by random parameters

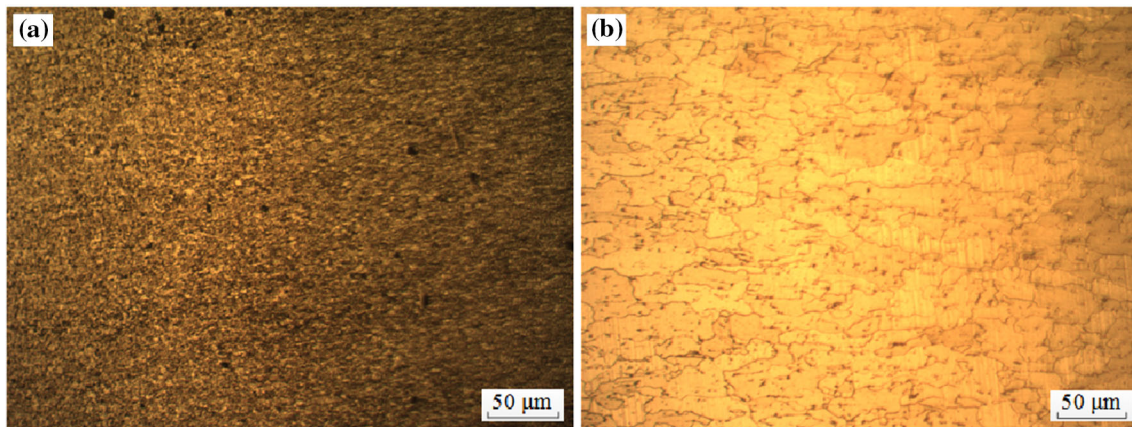


Fig. 14 BM microstructure. **a** The low carbon steel (Q235); **b** The stainless steel (SUS301L-HT)

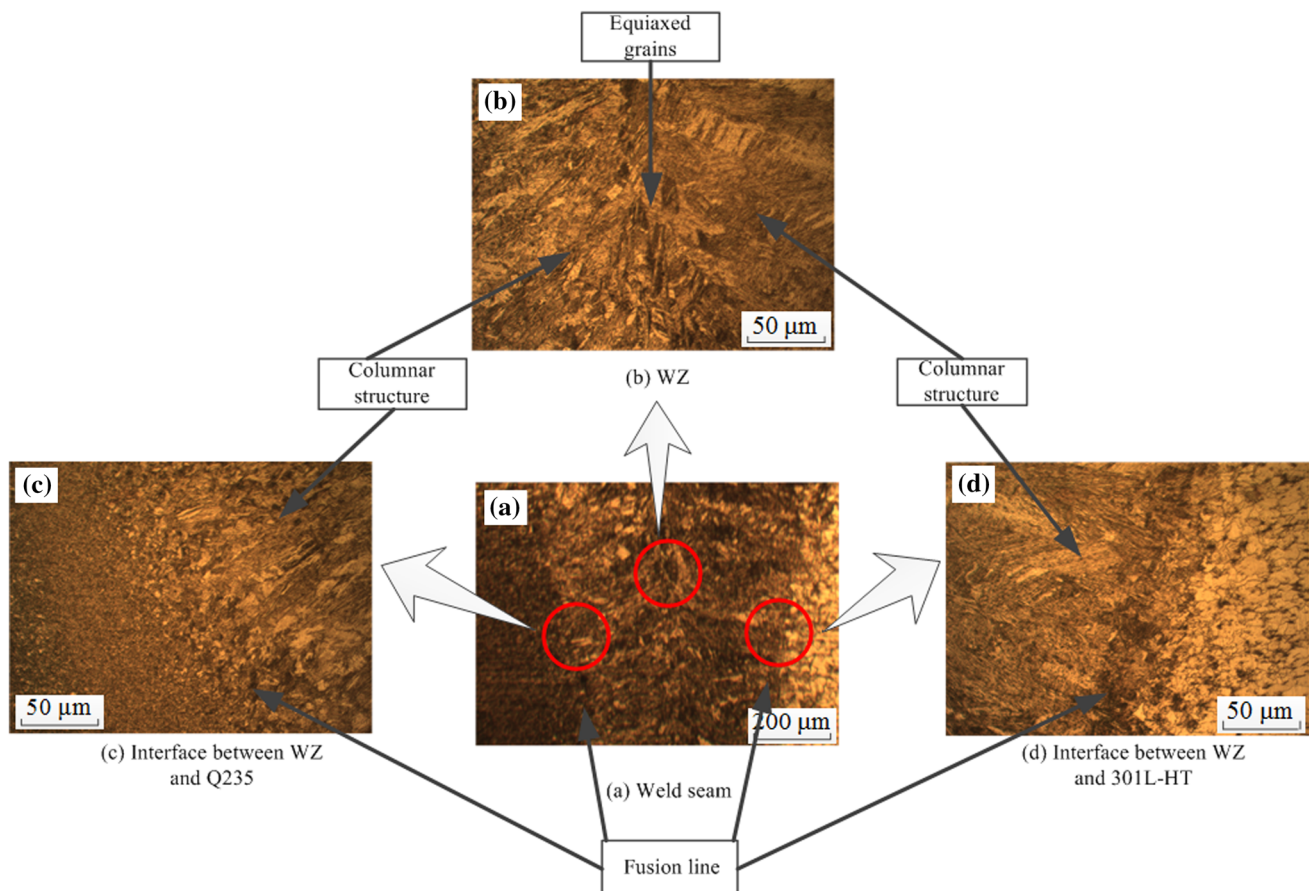


Fig. 15 Microstructures of WZ and interface between WZ and BM. **a** Weld seam; **b** WZ; **c** interface between WZ and Q235; **d** interface between WZ and 301L-HT

Figure 17 shows the tensile-tested results of the specimens from random parameters. Most of them are fractured in the weld seam. It shows that the tensile strength of the weld seam generated using random parameters is lower than that of the BM. Compared with the specimens (shown in Fig. 16) from optimized parameters, the

mechanical property of the unoptimized specimens is not reliable. It is clearly seen that the geometric optimization of LBW of dissimilar materials by GA is also beneficial regarding the mechanical properties. Thus, the optimization process has important practical utility for LBW of dissimilar materials.

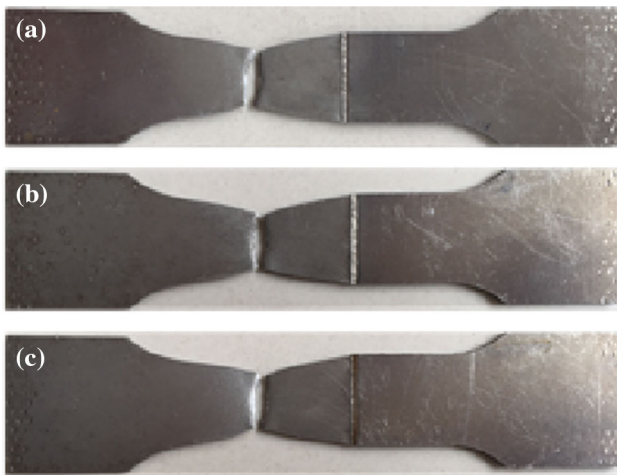


Fig. 16 Tensile-tested specimens generated by optimized parameters

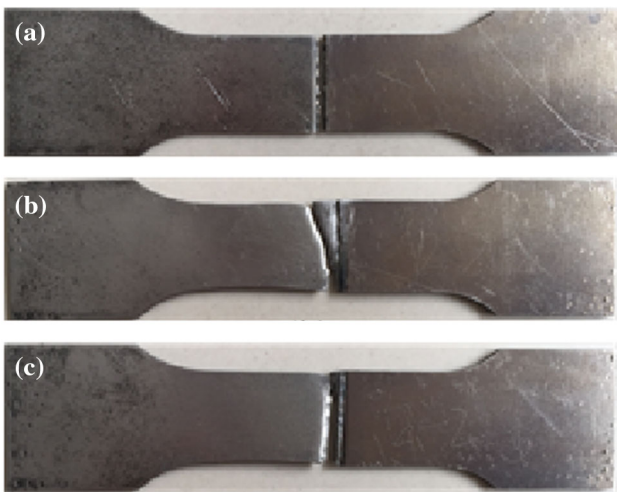


Fig. 17 Tensile-tested specimens generated by random parameters

5 Conclusion

In this paper, the process parameter modeling and optimization method of RBFNN and GA for LBW of dissimilar materials is proposed to obtain the desired welded joints. The whole LBW process using the corresponding specimens is implemented as the designed Taguchi experiments. The relationship between the input process parameters and the output responses of weld seam geometry is established by RBFNN. The predicted results from the RBFNN are embedded into the GA for the process parameters optimization. Then, confirmation experiment is conducted to validate the optimized parameters. The experimental results of weld seam geometry demonstrate good agreement with the optimal weld seam. Finally, the quality on the microstructures and mechanical property of the weld seam obtained from the optimal parameters are

further discussed. The microstructure is found to meet the requirement. The defects in the optimized weld seam are eliminated and the mechanical properties are improved compared to the unoptimized weld seam. Therefore, the proposed RBFNN-GA optimization method is effective and acceptable and can be used for guiding the practical production.

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